

■ 시나리오별 백업 복구(User managed)

KEY WORDS

특정한 데이터 파일 손상(백업이후에 리두정보가 있을 경우(완전복구가능)),
특정한 데이터파일이 손상(백업 이후에 리두정보가 없을 경우)(불완전복구)

■ 백업 대상 파일

Datafiles/Controlfiles/Redologfiles

SYS@orcl >select name from v\$datafile

```
2 union
3 select name from v$controlfile
4 union
5 select member from v$logfile;
```

NAME

```
-----
/u01/app/oracle/oradata/ORCL/control01.ctl
/u01/app/oracle/oradata/ORCL/control02.ctl
/u01/app/oracle/oradata/ORCL/redo01.log
/u01/app/oracle/oradata/ORCL/redo02.log
/u01/app/oracle/oradata/ORCL/redo03.log
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
/u01/app/oracle/oradata/ORCL/system01.dbf
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/users01.dbf
```

■ 컨트롤 파일 단일화

SYS@orcl> alter system set control_files = '/u01/app/oracle/oradata/ORCL/control01.ctl' scope = spfile;

SYS@orcl> show parameter control_files

NAME	TYPE	VALUE
control_files	string	/u01/app/oracle/oradata/ORCL/control01.ctl

■ 리두로그 파일(선택)

SYS@orcl >select * from v\$log;

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
1	1	97	10485760	512	1	NO
INACTIVE		2090995	02-MAY-25	2097189		
02-MAY-25		0				
2	1	98	10485760	512	1	NO
CURRENT		2097189	02-MAY-25	1.8447E+19		
		0				

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
1	1	97	10485760	512	1	NO
INACTIVE		2090995	02-MAY-25	2097189		
02-MAY-25		0				
2	1	98	10485760	512	1	NO
CURRENT		2097189	02-MAY-25	1.8447E+19		
		0				

```

-----
NEXT_TIME          CON_ID
-----
          3          1          96 10485760          512          1 NO
INACTIVE          2081984 01-MAY-25          2090995
02-MAY-25          0

```

0

■ 초기 파라미터파일 (선택)

SYS@orcl> show parameter spfile

NAME	TYPE	VALUE
spfile	string	/u01/app/oracle/product/19.3.0/ /dbhome_1/dbs/spfileorcl.ora

■ 아카이브 로그 모드 확인 방법

```

SYS@orcl> archive log list
Database log mode          No Archive Mode
Automatic archival         Disabled
Archive destination        /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch
Oldest online log sequence 96
Current log sequence       98

```

SYS@orcl> select name,log_mode from v\$database;

NAME	LOG_MODE
ORCL	NOARCHIVELOG
PWD	

SYS@orcl> !

```

[oracle@oel7v9r1 ~]$ mkdir -p backup/noarch
[oracle@oel7v9r1 ~]$ cd backup/noarch
[oracle@oel7v9r1 noarch]$ pwd
/home/oracle/backup/noarch
[oracle@oel7v9r1 noarch]$ exit
exit

```

SYS@orcl> create pfile='/home/oracle/backup/noarch/initorcl.ora' from spfile;

File created.

■ no archive mode 백업

- 일관성 있는 백업(close, cold, offline backup)
- 데이터베이스를 정상적으로 종료한 후 백업을 수행
 - shutdown normal | transactional | immediate
- whole database backup : 모든 데이터파일 컨트롤파일, 리두로그파일

```

SYS@orcl> select checkpoint_change# from v$database;
CHECKPOINT_CHANGE#

```

2097189

SYS@orcl> select scn_to_timestamp(checkpoint_change#) from v\$database;

SCN_TO_TIMESTAMP(CHECKPOINT_CHANGE#)

```
-----
02-MAY-25 02.08.31.000000000 AM
```

```
SYS@orcl> select name, checkpoint_change# from v$datafile;
```

```
NAME
```

```
-----
CHECKPOINT_CHANGE#
```

```
-----
/u01/app/oracle/oradata/ORCL/system01.dbf
2097189
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
2097189
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
2097189
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
2097189
```

```
-- 해당 scn번호(2097189) 이전의 트랜잭션은 복구작업이 필요 없음
```

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
1	1	97	10485760	512	1	NO
INACTIVE		2090995	02-MAY-25	2097189		
02-MAY-25		0				
2	1	98	10485760	512	1	NO
CURRENT		2097189	02-MAY-25	1.8447E+19		
		0				

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
3	1	96	10485760	512	1	NO
INACTIVE		2081984	01-MAY-25	2090995		
02-MAY-25		0				

```
SYS@ora19c> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
```

```
SYS@orcl> !
```

```
[oracle@oel7v9r1 ~]$ cd /u01/app/oracle/oradata/ORCL/
```

```
[oracle@oel7v9r1 ORCL]$ ls
```

```
control01.ctl redo01.log redo03.log system01.dbf undotbs01.dbf
control02.ctl redo02.log sysaux01.dbf temp01.dbf users01.dbf
```

```
[oracle@oracle19c ORA19C]$ ls /home/oracle/backup/noarch
initora19c.ora
```

```
[oracle@oel7v9r1 ORCL]$ cp -v *.{dbf,log,ctl} /home/oracle/backup/noarch/
'sysaux01.dbf' -> '/home/oracle/backup/noarch/sysaux01.dbf'
'system01.dbf' -> '/home/oracle/backup/noarch/system01.dbf'
'temp01.dbf' -> '/home/oracle/backup/noarch/temp01.dbf'
'undotbs01.dbf' -> '/home/oracle/backup/noarch/undotbs01.dbf'
'users01.dbf' -> '/home/oracle/backup/noarch/users01.dbf'
'redo01.log' -> '/home/oracle/backup/noarch/redo01.log'
'redo02.log' -> '/home/oracle/backup/noarch/redo02.log'
'redo03.log' -> '/home/oracle/backup/noarch/redo03.log'
'control01.ctl' -> '/home/oracle/backup/noarch/control01.ctl'
'control02.ctl' -> '/home/oracle/backup/noarch/control02.ctl'
```

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               754974720 bytes
Database Buffers            369098752 bytes
Redo Buffers                 7880704 bytes
Database mounted.
Database opened.
SYS@orcl> select checkpoint_change# from v$database;
```

```
CHECKPOINT_CHANGE#
-----
                2098460
```

```
SYS@orcl> select name, checkpoint_change# from v$datafile;
```

```
NAME
-----
CHECKPOINT_CHANGE#
-----
/u01/app/oracle/oradata/ORCL/system01.dbf
                2098460

/u01/app/oracle/oradata/ORCL/sysaux01.dbf
                2098460

/u01/app/oracle/oradata/ORCL/undotbs01.dbf
                2098460
```

```
NAME
-----
CHECKPOINT_CHANGE#
-----
/u01/app/oracle/oradata/ORCL/users01.dbf
                2098460
```

<<<시나리오 1>>> 특정한 데이터 파일 손상(백업이후에 리두정보가 있을 경우(완전복구가능))

```
<<hr>>
HR@orcl> create table hr.emp_temp(id number) tablespace users;
```

```
HR@orcl> insert into hr.emp_temp(id) values(1);
```

```
HR@orcl> commit;
Commit complete.
```

```
HR@orcl> select * from hr.emp_temp;
```

```

ID
-----
1

```

■ 장애발생

```

HR@orcl> conn /as sysdba
Connected.
SYS@orcl> shut immediate
Database closed.
Database dismounted.
ORACLE instance shut down.

```

```

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/users01.dbf

```

```

SYS@orcl> startup
ORACLE instance started.

```

```

Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               754974720 bytes
Database Buffers            369098752 bytes
Redo Buffers                 7880704 bytes
Database mounted.
ORA-01157: cannot identify/lock data file 7 - see DBWR trace file
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

```

```

SYS@orcl> !ls /u01/app/oracle/oradata/ORCL
control01.ctl redo01.log redo03.log      system01.dbf undotbs01.dbf
control02.ctl redo02.log sysaux01.dbf temp01.dbf

```

```

SYS@orcl> select status from v$instance;

```

```

STATUS
-----
MOUNTED

```

■ 복구 수행해야할 데이터파일 정보

```

SYS@orcl> select * from v$recover_file;

```

FILE#	ONLINE	ONLINE_	CHANGE#
7	ONLINE	ONLINE	0
FILE NOT FOUND			

```

SYS@orcl> select file#, name, status from v$datafile;

```

```

FILE#
-----
NAME
-----
STATUS
-----

```

```

1
/u01/app/oracle/oradata/ORCL/system01.dbf
SYSTEM

```

```

3
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
ONLINE

```

```

FILE#
-----
NAME
-----
STATUS
-----

```

```

4
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
ONLINE

```

```

7
/u01/app/oracle/oradata/ORCL/users01.dbf

```

```

FILE#
-----
NAME
-----
STATUS
-----
ONLINE

```

■ 복구

noarchive log mode에서는 특정한 데이터파일을 offline모드로 수행할 경우 꼭 offline drop으로 해야한다.

```
SYS@orcl> alter database datafile 7 offline drop;
```

Database altered.

```
SYS@orcl> alter database open;
```

Database altered.

가장 최근의 받은 백업본을 이용해서 restore

```

SYS@orcl> !
[oracle@oel7v9r1 ~]$ ls
Desktop  Downloads  Pictures  Templates  backup
Documents Music      Public    Videos    envora
[oracle@oel7v9r1 ~]$ cd backup/noarch/
[oracle@oel7v9r1 noarch]$ ls
control01.ctl redo01.log sysaux01.dbf undotbs01.dbf
control02.ctl redo02.log system01.dbf users01.dbf
initiorcl.ora redo03.log temp01.dbf
[oracle@oel7v9r1 noarch]$ cp -v users01.dbf /u01/app/oracle/oradata/ORCL/users01.dbf
'users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
[oracle@oel7v9r1 noarch]$ exit
exit

```

백업이후에 변경된 정보를 가지고 복구작업

```
SYS@orcl> recover datafile 7;
```

Media recovery complete.

```
SYS@orcl> select file#, name, status from v$datafile;
```

```
FILE#
-----
NAME
-----
STATUS
-----
      1
/u01/app/oracle/oradata/ORCL/system01.dbf
SYSTEM

      3
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
ONLINE
```

```
FILE#
-----
NAME
-----
STATUS
-----
      4
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
ONLINE

      7
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
FILE#
-----
NAME
-----
STATUS
-----
OFFLINE
```

```
SYS@orcl> alter database datafile 7 online;
```

```
Database altered.
```

```
SYS@orcl> select file#, name, status from v$datafile;
```

```
FILE#
-----
NAME
-----
STATUS
-----
      1
/u01/app/oracle/oradata/ORCL/system01.dbf
SYSTEM

      3
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
ONLINE
```

```
FILE#
-----
NAME
-----
STATUS
-----
```

```

4
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
ONLINE

```

```

7
/u01/app/oracle/oradata/ORCL/users01.dbf

```

```

FILE#
-----
NAME
-----
STATUS
-----
ONLINE

```

```
SYS@orcl> select * from hr.emp_temp;
```

```

ID
-----
1

```

■ 정상적으로 올라왔을 경우 shutdown immediate 후 /u01/app/oracle/oradata/ORCL/*.{dbf,log,ctl}을
이용해서 백업본 새로 만들어 두기

```

SYS@orcl> !
[oracle@oel7v9r1 ~]$ cd backup/noarch/
[oracle@oel7v9r1 noarch]$ ls
control01.ctl redo01.log sysaux01.dbf undotbs01.dbf
control02.ctl redo02.log system01.dbf users01.dbf
initiorcl.ora redo03.log temp01.dbf
[oracle@oel7v9r1 noarch]$ rm -rf *
[oracle@oel7v9r1 noarch]$ ls
[oracle@oel7v9r1 noarch]$ cp -v /u01/app/oracle/oradata/ORCL/*.{dbf,log,ctl} .
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> './sysaux01.dbf'
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> './system01.dbf'
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> './temp01.dbf'
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> './undotbs01.dbf'
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> './users01.dbf'
'/u01/app/oracle/oradata/ORCL/redo01.log' -> './redo01.log'
'/u01/app/oracle/oradata/ORCL/redo02.log' -> './redo02.log'
'/u01/app/oracle/oradata/ORCL/redo03.log' -> './redo03.log'
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> './control01.ctl'
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> './control02.ctl'
[oracle@oel7v9r1 noarch]$ ls
control01.ctl redo01.log redo03.log system01.dbf undotbs01.dbf
control02.ctl redo02.log sysaux01.dbf temp01.dbf users01.dbf
[oracle@oel7v9r1 noarch]$ pwd
/home/oracle/backup/noarch
[oracle@oel7v9r1 noarch]$ exit
exit

```

```
SYS@orcl> create pfile='/home/oracle/backup/noarch/initiorcl.ora' from spfile;
```

```
File created.
```


<<<시나리오 2>>> 특정한 데이터파일이 손상(백업 이후에 리두정보가 없을 경우)

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               754974720 bytes
Database Buffers           369098752 bytes
Redo Buffers                7880704 bytes
Database mounted.
Database opened.
SYS@orcl> select checkpoint_change# from v$database;
```

```
CHECKPOINT_CHANGE#
```

```
-----
2102516
```

```
SYS@orcl> select name, checkpoint_change#, status from v$datafile;
```

```
NAME
```

```
-----
CHECKPOINT_CHANGE# STATUS
```

```
-----
/u01/app/oracle/oradata/ORCL/system01.dbf
2102516 SYSTEM
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
2102516 ONLINE
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
2102516 ONLINE
```

```
NAME
```

```
-----
CHECKPOINT_CHANGE# STATUS
```

```
-----
/u01/app/oracle/oradata/ORCL/users01.dbf
2102516 ONLINE
```

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC

STATUS	FIRST_CHANGE#	FIRST_TIME	NEXT_CHANGE#			

NEXT_TIME	CON_ID					

1	1	97	10485760	512	1	NO
INACTIVE		2090995	02-MAY-25	2097189		
02-MAY-25		0				
2	1	98	10485760	512	1	NO
CURRENT		2097189	02-MAY-25	1.8447E+19		
		0				

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC

STATUS	FIRST_CHANGE#	FIRST_TIME	NEXT_CHANGE#			

NEXT_TIME	CON_ID					

```

          3          1          96  10485760          512          1 NO
INACTIVE          2081984 01-MAY-25          2090995
02-MAY-25

```

■DB변화

SYS@orcl> select * from hr.emp_temp;

```

          ID
-----
          1

```

SYS@orcl> insert into hr.emp_temp(id) values(2);

1 row created.

SYS@orcl> commit;

Commit complete.

SYS@orcl> select * from hr.emp_temp;

```

          ID
-----
          1
          2

```

```

SYS@orcl> select f.file_name
2  from dba_extents e, dba_data_files f
3  where e.file_id = f.file_id
4  and e.segment_name = 'EMP_TEMP'

```

FILE_NAME

```

-----
/u01/app/oracle/oradata/ORCL/users01.dbf

```

SYS@orcl> alter system switch logfile; ➔ 리두로그 over write.

System altered.

SYS@orcl> /

System altered.

SYS@orcl> /

System altered.

SYS@orcl> /

System altered.

SYS@orcl> select * from v\$log;

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
1	1	100	10485760	512	1	NO
INACTIVE		2103440	02-MAY-25	2103443		
02-MAY-25		0				
2	1	101	10485760	512	1	NO

```

INACTIVE          2103443 02-MAY-25          2103450
02-MAY-25          0

```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC
3	1	102	10485760	512	1	NO
CURRENT		2103450	02-MAY-25	1.8447E+19		

-- SEQUENCE# 번호 변경(98->102)된 것 확인

■ 장애 발생

```

SYS@orcl> shut immediate
Database closed.
Database dismounted.
ORACLE instance shut down.

```

```

SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/users01.dbf
-rw-r-----. 1 oracle dba 5251072 May  2 03:17 /u01/app/oracle/oradata/ORCL/users01.dbf

```

```

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/users01.dbf

```

```

SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/users01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/users01.dbf: No such file or directory

```

■ 장애 확인

```

SYS@orcl> startup
ORACLE instance started.

```

```

Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size              754974720 bytes
Database Buffers          369098752 bytes
Redo Buffers                7880704 bytes
Database mounted.
ORA-01157: cannot identify/lock data file 7 - see DBWR trace file
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

```

```

SYS@orcl> select * from v$recover_file;

```

FILE#	ONLINE	ONLINE_	CHANGE#
7	ONLINE	ONLINE	0
FILE NOT FOUND			

```

SYS@orcl> select name, checkpoint_change#, status from v$datafile;

```

NAME	CHECKPOINT_CHANGE#	STATUS

```
-----  
/u01/app/oracle/oradata/ORCL/system01.dbf  
2103745 SYSTEM
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf  
2103745 ONLINE
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf  
2103745 ONLINE
```

NAME

CHECKPOINT_CHANGE# STATUS

```
-----  
/u01/app/oracle/oradata/ORCL/users01.dbf  
2103745 ONLINE
```

■복구시행

```
SYS@orcl> alter database datafile 7 offline drop;
```

Database altered.

```
SYS@orcl> select file#, name, status from v$datafile;
```

```
      FILE#  
-----  
NAME  
-----  
STATUS  
-----  
          1  
/u01/app/oracle/oradata/ORCL/system01.dbf  
SYSTEM  
  
          3  
/u01/app/oracle/oradata/ORCL/sysaux01.dbf  
ONLINE
```

```
      FILE#  
-----  
NAME  
-----  
STATUS  
-----  
  
          4  
/u01/app/oracle/oradata/ORCL/undotbs01.dbf  
ONLINE
```

```
          7  
/u01/app/oracle/oradata/ORCL/users01.dbf  
  
      FILE#  
-----  
NAME  
-----  
STATUS  
-----  
OFFLINE
```

```
SYS@orcl> alter database open;
```

Database altered.

```
SYS@orcl> select status from v$instance;
```

```
STATUS
-----
OPEN
```

```
SYS@orcl> select count(*) from hr.employees;    ==> 질의 가능
```

```
      COUNT(*)
-----
         107
```

```
SYS@orcl> select * from hr.emp_temp;           ==> 질의 불가능
select * from hr.emp_temp
                *
```

```
ERROR at line 1:
ORA-00376: file 7 cannot be read at this time
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

■ 복구 수행해야할 백업 데이터 파일 찾아서 restore(복사 명령어 cp)

```
SYS@ora19c> !
[oracle@oracle19c ~]$ cd backup/noarch/
[oracle@oracle19c noarch]$ ls
control01.ctl  initora19c.ora  redo05.log  redo06.log  redo07.log  sysaux01.dbf  system01.dbf  temp01.dbf
undotbs01.dbf  users01.dbf
[oracle@oracle19c noarch]$ cp -v users01.dbf /u01/app/oracle/oradata/ORCL/users01.dbf
'users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

no archive log mode에서 로그스위치가 발생하면 리두 정보가 overwrite가 되어버리기때문에 완전복구를 할 수 없음!

```
SYS@ora19c> recover datafile 7
ORA-00279: change 3266072 generated at 02/12/2025 11:56:25 needed for thread 1
ORA-00289: suggestion :
/u01/app/oracle/fast_recovery_area/ORCL/archivelog/2025_02_12/o1_mf_1_270_%u_.arc
ORA-00280: change 3266072 for thread 1 is in sequence #270
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
ORA-00308: cannot open archived log
'/u01/app/oracle/fast_recovery_area/ORCL/archivelog/2025_02_12/o1_mf_1_270_%u_.arc'
ORA-27037: unable to obtain file status
SYS@orcl> !cp -v backup/noarch/users01.dbf /u01/app/oracle/oradata/ORCL/users01.dbf
'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> recover datafile 7;
ORA-00279: change 2102513 generated at 05/02/2025 03:04:20 needed for thread 1
ORA-00289: suggestion :
/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_98_1199933387.dbf
ORA-00280: change 2102513 for thread 1 is in sequence #98
```

```
Specify log: {<RET>=suggested | filename | AUTO | CANCEL}
auto
```

```
ORA-00308: cannot open archived log
'/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_98_1199933387.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
```

```
ORA-00308: cannot open archived log
'/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_98_1199933387.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
```

■ 완전복구가 실패하였다 리두의 시퀀스 번호가 다르기 때문에
no archive log mode에서 완전복구를 할 수 없으면 컨트롤파일, 데이터파일, 리두로그파일 전부를 restore.
즉, 마지막 백업시점으로 데이터베이스를 복구한다.

```
SYS@orcl> shut immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
SYS@orcl> !
[oracle@oel7v9r1 ~]$ cd backup/noarch/
[oracle@oel7v9r1 noarch]$ ls
control01.ctl redo01.log sysaux01.dbf undotbs01.dbf
control02.ctl redo02.log system01.dbf users01.dbf
initiorcl.ora redo03.log temp01.dbf
[oracle@oel7v9r1 noarch]$ cp -v *.{dbf,log,ctl} /u01/app/oracle/oradata/ORCL/
'sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
'redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
'redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'
'redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
'control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
[oracle@oel7v9r1 noarch]$ exit
exit
```

```
SYS@orcl> startup
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               754974720 bytes
Database Buffers            369098752 bytes
Redo Buffers                 7880704 bytes
Database mounted.
Database opened.
SYS@orcl> select * from hr.emp_temp;
```

```
      ID
-----
      1
```

```
SYS@orcl>
```